

Measurement of exchange rates

There are a number of ways of measuring the price of a currency. It can, for example, be measured in terms of how much it costs in another currency. It can also be measured in terms of what it can buy in another country or countries and in terms of a basket of currencies.

Nominal and real exchange rates

The nominal foreign exchange rate is the price of one currency in terms of another currency. For example, 160 Pakistani rupees may buy one US dollar.

Economists also measure real exchange rates. A real exchange rate takes price changes as well as exchange rate changes into account to assess changes in the competitiveness of a country's products in global markets. A fall in a country's foreign exchange rate would be expected to make its exports more price competitive. However, if the country is experiencing a relatively high inflation rate, export prices may actually be increasing. A real exchange rate compares the relative price of two countries' products:

Real exchange rate = nominal exchange rate $\times$ domestic price index/foreign price index

The real exchange rate may rise as a result of an appreciation or an increase in the country's relative inflation rate. The price of products in the country will increase relative to those in another country or countries. More imports could be purchased for a given quantity of exports. As the country's products are more expensive relative to its competitors, the country is likely to import more and export less. The real exchange rate has more of an impact on the current account balance than the nominal exchange rate as it reflects international price competitiveness.

Trade-weighted exchange rate

In practice, a country trades with a number of other countries and it is possible that its currency may rise in value against some countries while falling against others. To gain an impression of the general change in a country's foreign exchange rate, economists measure trade-weighted exchange rates. A trade-weighted exchange rate is a measure, in index form, of the price of a currency against a basket of currencies. These are weighted according to the relative importance of the countries in the country's trade. For example, if India undertakes three times as much trade with China as it does with the US, the Chinese renminbi will be given three times as much weight in the calculation as the US dollar.

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Determination of exchange rates

Exchange rates can be determined by a government, by the free market forces of demand and supply or by a combination of the two. Most governments and their central banks today allow market forces to play a large role in determining their countries' exchange rates, but may intervene if there are large fluctuations in their value.

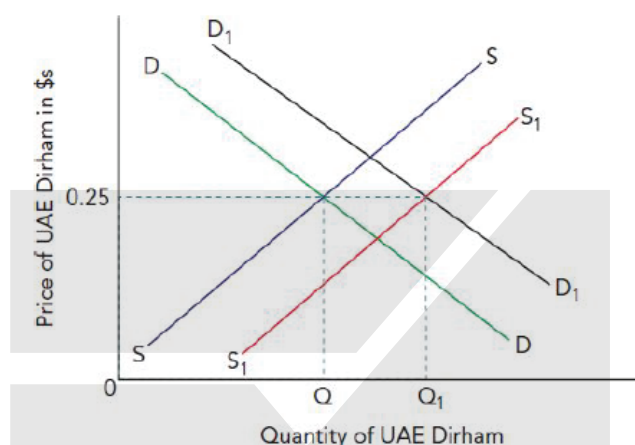


Figure 49.4: Central bank action to maintain the value of the UAE dirham

A fixed exchange rate system

In a fixed exchange rate system, the price of the currency is determined by the government. For instance, the value of the UAE's dirham may be fixed at 1 dirham = US\$0.25. The UAE's central bank will maintain the rate by direct intervention and/or by trying to influence the market demand for and supply of the currency. Direct intervention will involve the central bank buying or selling the currency. It may try to influence the market demand and supply by changing the rate of interest. If, for instance, there is downward pressure on the exchange rate because the supply of the currency is increasing on the foreign exchange market, the central bank is likely to buy the currency. The central bank may also increase the rate of interest. Such a measure may attract hot money flows, with people buying the currency to place into accounts in the country's financial institutions. Figure 49.4 shows that the UAE government fixes its exchange rate at 1 dirham = \$0.25.

If the supply of the currency increases to S_1 , the central bank may buy enough of the currency to shift the demand curve to D_1 , and maintain the value of the currency at \$0.25.

It is easier for a central bank to maintain a fixed exchange rate if that rate is set close to the long-run equilibrium value of the currency. This is because it will only have to offset short-run – and possibly relatively small – fluctuations. If, however, the exchange rate is overvalued, the central bank may be in danger of running out of reserves trying to keep the exchange rate at a level that does not reflect its market price.

One of the disadvantages of a fixed exchange rate is the need to keep reserves. This involves an opportunity cost, as the foreign exchange or gold, for instance, could be used for other purposes.

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There is also the risk that a government may sacrifice other policy objectives in order to maintain the fixed exchange rate. For instance, a central bank may raise the interest rate in a bid to keep the exchange rate at the price the government wants maintained. However, a higher interest rate may reduce aggregate demand and increase unemployment.

Market pressures may mean that the rate at which the currency is fixed may have to be changed over time. A reduction in the value of a fixed exchange rate to a lower level is known as devaluation. A revaluation occurs when the government raises the exchange rate to a new, higher fixed rate.

A fixed exchange rate does create certainty, which can promote international trade and investment. For instance, a foreign firm may be more willing to set up a branch in the country if it knows how much any revenue it earns will be worth in its own currency.

A fixed exchange rate also imposes discipline on a government to keep inflation low, so that a loss of international price competitiveness does not put downward pressure on the exchange rate.

A managed system

A managed system, in effect, combines features of a floating exchange rate system and a fixed exchange rate system. It usually involves a government allowing the exchange rate to be determined by market forces within a given band, which has upper and lower limits. Figure 49.5 shows that the government sets a central value of P , an upper band of P_a , which it does not want the exchange rate to exceed, and a lower limit of P_b , which it does not want it to fall below.

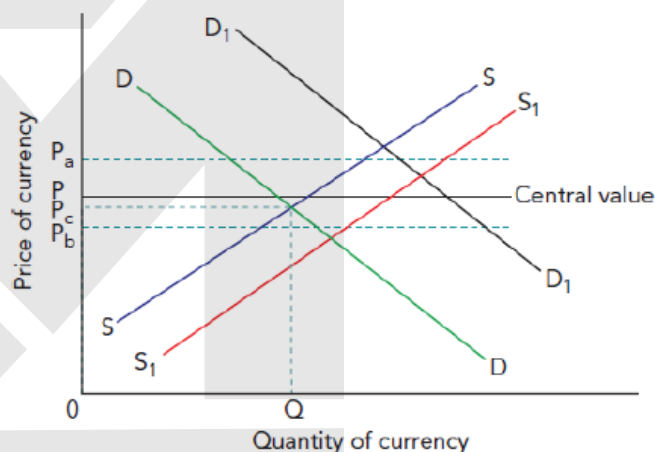


Figure 49.5: A managed system

If the exchange rate is within the limits, for example at P_c , no action will be taken. If, however, demand for the currency were to rise to D_1 , the central bank would sell the domestic currency to increase supply to S_1 and so keep the exchange rate within the desired band.

Revaluation and devaluation of a fixed exchange rate

A revaluation occurs when a government raises the exchange rate to a new, higher fixed rate. This higher price will be maintained by the government. If there is downward pressure on the currency, the central bank will buy the currency, using some of its reserves. It might also raise the interest rate.

A devaluation is a reduction in the price of the currency arising from a government's decision to maintain it at a new lower level. If there is upward pressure on the new, lower exchange rate, the central bank will sell its currency. The increased supply of the currency on the foreign exchange market may keep the price low. The central bank might also lower the interest rate to reduce demand for the currency and increase its supply.

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Devaluations and revaluations occur for two main reasons. One reason is that the current exchange rate is not sustainable. It results in surpluses or shortages of the currency in the foreign exchange market. The second reason is the use of devaluations and revaluations as a policy tool.

Changes in the exchange rate under different exchange rate systems

Section 28.4 discussed how a floating exchange rate will change when there are changes in the market forces of demand and supply. The main reasons why demand for the currency may rise are increases in demand for the country's goods and services, increases in direct and portfolio investment in the country and speculation that the currency will rise in price. The supply of the currency will decrease if the country buys fewer imports, invests less abroad and it is expected that the currency will rise in price.

A fixed exchange rate does not often change. A change when it does occur is the result of a government decision. As mentioned above, a government will devalue its currency if the government thinks that it cannot maintain its current price. If market forces are putting downward pressure on the price of the currency, there is a risk that the central bank will run out of the reserves that it can use to buy the currency. A government may also be reluctant for its central bank to raise the interest rate to encourage hot money flows. This is because the higher interest rate could reduce economic activity. A government may also devalue the currency to increase the international competitiveness of its products so as to reduce a current account deficit, increase economic growth and reduce unemployment.

A government may instruct its central bank to bring about a revaluation of the currency if it thinks there is too much upward pressure on the currency. The government may not want to sell large quantities of its currency to keep the exchange rate down to its target rate as this action may add to the money supply. A government may also revalue its currency to reduce inflationary pressure. A higher exchange rate will reduce the price of imports, which can increase the pressure on domestic industries to become more internationally competitive.

The effects of changing exchange rates on the external economy

The effect of a change in the exchange rate on the current account balance will be influenced by the price elasticities of demand for exports and imports (see Section 28.1). The requirement for the combined elasticities to exceed 1 for the trade balance, and so the current account balance, to be improved by a change in the exchange rate is known as the Marshall-Lerner condition. If this condition is met, a fall in the exchange rate would reduce a current account deficit, while a rise in the exchange rate would reduce a current account surplus. The table below uses simplified figures to illustrate the Marshall-Lerner condition in the case of a fall in the exchange rate.

The greater the combined PED for exports and imports, the smaller will be the fall in the exchange rate required to improve the current account position. If the PED is less than 1, a revaluation of the exchange rate would be the more appropriate policy strategy.

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Initially £1 (UK currency) = \$1.5

- 40 products, which would sell at £10 each in the home market, sell for \$15 in the US. Export revenue = £400 (\$600)
- 50 imports from the US, which sell in the US at \$30, are purchased in the UK at £20 each. Import expenditure = £1000 (\$1500)
- Deficit = £400 – £1000 = £600

Then the pound sterling falls in value to £1 = \$1.

- PED for exports = 0. Price in US falls to \$10 each. Total export revenue is now $40 \times \$10 = \400 which is now converted into £400, leaving export revenue in pounds sterling unchanged.
- PED for imports = -1.2. Price of the \$30 product will rise to £30, a 50% increase. Demand falls by 60% to 20. Total import expenditure now equals £600.
- The deficit is now £400 – £600 = £200. It has fallen by £400.

The J curve effect is related to the Marshall-Lerner condition. In some cases, a fall in the exchange rate will actually worsen the current account position before it starts to improve it. This is because, in the short run, demand for imports and exports may be relatively inelastic. It takes time to recognize that prices have changed and then to search for alternative products. In the long run, demand becomes more elastic and the current account position may move from deficit into surplus as shown in Figure 49.9.

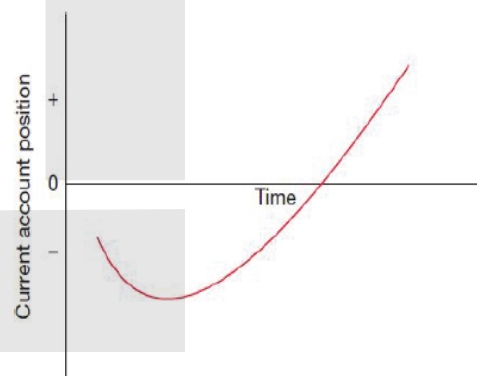


Figure 49.9: The J curve effect

A higher exchange rate may increase a current account deficit or reduce a current account surplus, but the outcome will depend mainly on the price elasticities of demand for exports and imports. The Marshall-Lerner condition and the J curve work in reverse. So, a current account surplus will only be reduced if the sum of the price elasticities of demand for exports and imports is greater than 1. A rise in the exchange rate may increase a current account surplus in the short run before reducing it in the longer run as shown in Figure 49.10.

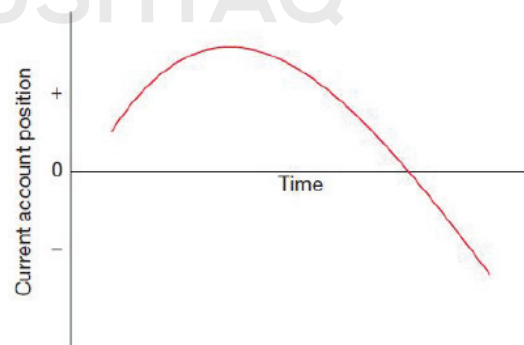


Figure 49.10: The reverse J curve effect

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